# Khalil Sahli

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*Machine Learning Engineer (GenAI, LLM agents, edge AI)*

## Education

**The National Engineering School of Carthage (Enicarthage)** — Charguia 2, Tunis
*Computer Engineering Degree – 3-year engineering cycle, Master's-equivalent* — *September 2024 – May 2027 (expected)*

**Tunis Preparatory Engineering Institute (IPEIT)** — Montfleury, Tunis
*Preparatory Classes for Engineering Schools – 2 years, Mathematics & Physics* — *September 2022 – May 2024*

## Experience

**Generative AI Engineer (Intern)** — June 2026 – August 2026
*STMicroelectronics* — *El Ghazela, Tunis*
- Designed the chunking and retrieval structure feeding an internal RAG system, then benchmarked it on 50 questions across 25 errata sheets, comparing flat BM25, metadata-scoped BM25 and structured search
- Built eight Python packages that turn technical PDFs into structured JSON, recovering 178 tables and 930 sections
- Guarded the extraction rules with 928 automated tests and a citation check that reopens each cited page

**Software Engineer (Intern)** — July 2025 – August 2025
*Sagemcom* — *Ez Zahra, Tunis*
- Cut manual remediation effort by 80% by automating Wi-Fi card corrections with idempotent update logic
- Built a Python tool that parses device-enclosure logs against a canonical reference and emits CSV reports
- Kept releases reliable across heterogeneous hardware by adding unit tests and CI pipelines with pytest

**Machine Learning Engineer (Student Researcher)** — October 2024 – November 2024
*IEEE Computer Tunisia Chapter* — *Soukra, Tunis*
- Held object recognition accuracy steady across changing lighting and scene conditions by building a CNN perception pipeline for real-time detection and classification from camera input, backed by a data augmentation stage
- Benchmarked detection architectures and loss functions to maximize mAP and inference speed on GPU hardware
- Cut model iteration time by scripting reproducible training and evaluation workflows in PyTorch and OpenCV

## Projects

**TicketPilot – AI Agent with RAG & Tool Calling** | *Python, LangGraph, Qdrant, FastAPI* — Apr 2026 – May 2026
- Built an AI agent that classifies each request, retrieves the applicable rule, calls business tools through REST APIs and drafts a cited answer: 45% of 150 labeled cases resolved automatically at 97.1% precision, the rest routed to human review
- Cut unsupported claims from 11.3% to 2.7% with a verification node; hard-coded guardrails blocked 12/12 prompt injections
- Lifted recall@5 from 72.0% (BM25) and 78.7% (embeddings) to 86.0% with hybrid vector search (RRF) in Qdrant

**SpiceFlow – Natural Language to Circuit** | *Python, C++, CodeLlama-13B, LoRA, ngspice* — Jan 2026 – May 2026
- Fine-tuned CodeLlama-13B with LoRA (rank 128, bf16) on 5,737 description-netlist pairs to write SPICE netlists from plain English: 78% valid over 250 test prompts, 92% on passive filters
- Drew schematics with a custom C++ engine (OGDF layout to SVG, LTspice export): 97% of netlists rendered, 0.4 s avg.
- Simulated the circuits with ngspice (PySpice) behind a FastAPI backend with conversation memory and a React frontend

**CircuitVision – Detection, Segmentation & VLM** | *Python, PyTorch, YOLOv11, SAM 2, Docker* — Jan 2026 – Feb 2026
- Turned photos of circuit diagrams into simulation-ready SPICE netlists at 0.93 mAP-50, by fine-tuning YOLOv11 for component detection and chaining it with SAM 2 segmentation and a vision-language model that reads component labels
- Reached 98.7% segmentation accuracy on 267 annotated images by adapting SAM 2
- Removed local GPU dependency by containerizing the pipeline with Docker and serving it from RunPod

**TinyML CAN Bus Intrusion Detection on STM32** | *C, Python, TensorFlow Lite, STM32F446* — Sep 2026 – Nov 2026
- Built a real-time CAN intrusion detector on an STM32F446: rules plus a 1,441-parameter int8 neural network hand-written in C, 63 µs per frame, bit-exact with Python on 37.7M frames
- Flagged 99.2% of attack frames on held-out real-vehicle attacks (ROAD) and detected 10 of 11 attacks live on a 500 kbit/s bus; retrained for a second car in 5 minutes: 2.5 false warnings per hour
- Hardened the device: ECDSA secure boot, read/write protection, tamper erase, signed alerts, SecOC (AES-CMAC)

**VHDL Neural Network – FPGA Digit Recognition** | *VHDL, PyTorch, ModelSim, Spartan-6* — Sep 2026 – Oct 2026
- Ran a 64-32-10 MLP entirely in hardware: trained it in PyTorch to 97.50% test accuracy, quantized every weight to 8-bit Q2.6 with no loss against the float model, and derived the accumulator widths (15 and 19 bits) that make overflow impossible
- Built the datapath from gates in VHDL (array multipliers, a 6-level adder tree, ReLU shift-saturate, argmax tree) and verified it bit-exact against the Python reference on all 360 test images
- Deployed a MAC-based variant on a Nexys 3 Spartan-6 board, driven over USB-UART from a PC sketchpad

## Technical Skills

**Generative AI & LLMs**: RAG, AI agents, LangGraph, LoRA fine-tuning, Hugging Face, Qdrant, Gemini API, Whisper
**Machine Learning & Vision**: PyTorch, TensorFlow, scikit-learn, YOLOv11, SAM 2, OpenCV, model evaluation
**Edge AI & Embedded**: int8 quantization, TensorFlow Lite, STM32, CAN, secure boot, FreeRTOS, C, VHDL, FPGA
**Software & Tools**: Python, C++, SQL, FastAPI, Docker, Git, CI/CD, pytest, Linux, Windows, GCP
**Languages**: Arabic (native), English (C1), French (B2), Spanish (basic), Japanese (basic)